

MASTERCLASS

CrewAI

Multi-Agent AI Systems

Part-1 — Foundations & First Multi-Agent System

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Free & Open Source



Python 3.11



LLaMA 3 via Groq



Windows / Mac / Linux

Part-1 Agenda

01

Introduction to Agentic AI

What is agentic AI and why it matters now

02

Prompt AI vs Multi-Agent Systems

Key differences and when to use each approach

03

Core Concepts

Agents · Tasks · Crew · Process — the 4 building blocks

04

Role, Goal & Backstory

How to define agents that produce expert-level output

05

Sequential Workflow

How agents pass work to each other automatically

06

Hands-On Build

Build and run your first 3-agent content system

What is Agentic AI?

"An AI that can PLAN, REASON, ACT and ADAPT — not just respond."



Traditional AI

- ▶ You prompt → It responds
- ▶ One shot, one answer
- ▶ Stateless — forgets everything
- ▶ You do all the work



Agentic AI

- ▶ You give a goal → It figures out how
- ▶ Multi-step reasoning & action
- ▶ Uses tools, APIs, other agents
- ▶ Works autonomously toward the goal



Multi-Agent AI

- ▶ Multiple specialized agents
- ▶ Each agent has a role & expertise
- ▶ Agents collaborate like a team
- ▶ Produces far better results

Prompt-Based AI vs Multi-Agent Systems



Prompt-Based AI

- ▶ Single model, single response
- ▶ Human drives every step manually
- ▶ Stateless — no memory
- ▶ Limited by one context window
- ▶ Good for simple isolated tasks
- ▶ Example: "Summarize this text"

VS



Multi-Agent System

- ✓ Multiple specialized agents collaborating
- ✓ System works autonomously end-to-end
- ✓ Stateful — agents share context & memory
- ✓ Distributed reasoning across agents
- ✓ Built for complex multi-step workflows
- ✓ Example: Research → Analyze → Write → Review

4 Core Concepts of CrewAI



01

AGENT

An autonomous AI team member with a specific identity and expertise

Has: Role • Goal • Backstory • Tools • LLM

💡 Think of it as: A virtual employee you hire for a specific job



02

TASK

A specific unit of work assigned to an agent with clear deliverables

Has: Description • Expected Output • Agent • Context

💡 Think of it as: The job brief you give your employee



03

CREW

The team — a collection of agents and tasks working together

Has: Agents List • Tasks List • Process Type

💡 Think of it as: Your project team assembled for one mission



04

PROCESS

The workflow style that defines how agents collaborate

Has: Sequential (in order) • Hierarchical (manager assigns)

💡 Think of it as: Your project management methodology

Role · Goal · Backstory — The Secret Weapon

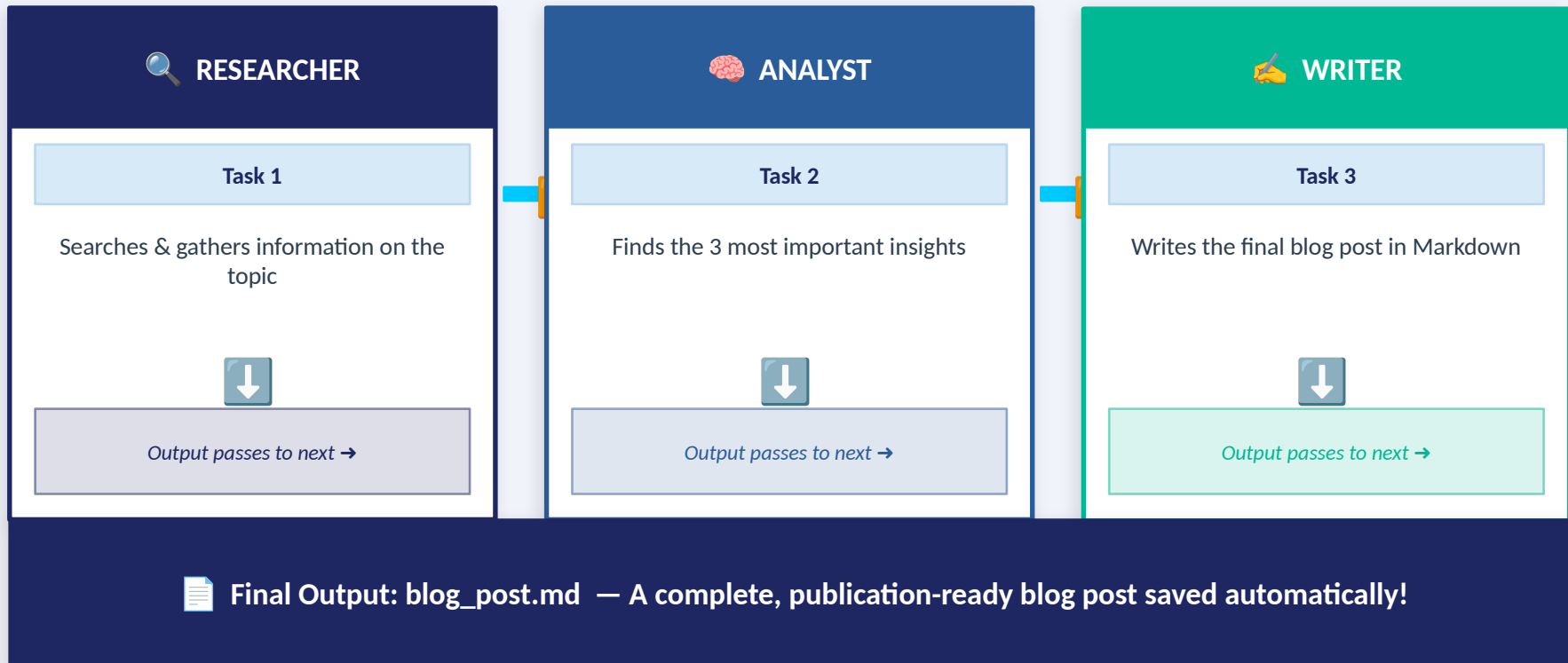
These 3 fields become the agent's system prompt. The more specific they are, the better the output.

	✗ Weak Definition	✓ Strong Definition
ROLE	Assistant	B2B SaaS Market Research Specialist
GOAL	Help with research	Identify market gaps in the CRM software space a startup can exploit in 2025
BACKSTORY	You are helpful	You have 10 years analyzing SaaS markets for VC firms. Known for synthesizing complex data into clear investment theses. Always back claims with data.

💡 *Pro Tip: Write backstories the way a novelist writes characters — "You are known for...", "Your approach is..."*

Sequential Workflow Architecture

Tasks run one after another. Each agent's output automatically becomes the next agent's input.



What Are Tools?

Tools are what turn AI agents from text generators into real-world actors.

Agent **WITHOUT** Tools

Can only use knowledge from training data

Limited to what it already knows

Cannot access live or real-time information

Agent **WITH** Tools

Can search the internet in real time

Can read files, call APIs, run code

Actions in the real world!

Common Tools Available in CrewAI



SerperDevTool

Google web search



WebsiteSearchTool

Search specific websites



FileReadTool

Read local files



CodeInterpreterTool

Execute Python code



ScrapeWebsiteTool

Scrape any URL



GithubSearchTool

Search GitHub repos



Hands-On: What We Will Build



Content Research & Writing Crew — 3 Agents • 3 Tasks • 1 Topic • Automatic Blog Post



Researcher

Tech Research Specialist

Searches the web for latest trends, statistics & expert opinions on the topic

Uses: SerperDevTool (web search)



Analyst

Strategic Content Analyst

Reads research findings and extracts the 3 most valuable insights for the audience

Uses: LLaMA 3 reasoning only



Writer

Senior Content Writer

Writes a complete 600-800 word Markdown blog post with intro, sections & CTA

Saves to: blog_post.md

INPUT: crew.kickoff(inputs={"topic": "Any Topic You Choose"}) → OUTPUT: blog_post.md ✓

Part-1 — Key Takeaways

1 Agentic AI = autonomous systems that plan, act and adapt — not just respond to prompts

2 CrewAI has 4 building blocks: Agent · Task · Crew · Process — master these and you master the framework

3 Agent quality = Role + Goal + Backstory quality. Be specific, be a novelist with your backstory!

4 Sequential Process: tasks run in order, each output automatically feeds the next agent

5 Tools give agents superpowers — web search, file access, APIs, code execution

6 You can build a production-useful multi-agent system in under 60 lines of Python — for FREE



Mini Assignment

Complete before Part-2 →



Your Challenge

Add a 4th Agent to today's Content Crew:



Agent: SEO Optimizer



Role: Reviews the blog post and suggests keyword & SEO improvements



Task: Set context=[writing_task] so it reads the finished blog post



Goal: Run the updated 4-agent crew and share your output!

Coming Up — Part-2:

Sequential vs Hierarchical Workflows · Multi-Agent Collaboration Patterns · Tools & APIs · Memory · Real-World Automation System



You Did It!

Your first multi-agent AI system is running.



3 Agents Built



3 Tasks Defined



Crew Running



Blog Post Saved

See you on Part-2! 

Sequential vs Hierarchical · Tools & APIs · Memory · Real-World Automation